

GRAND RAPIDS, MN, USA

WMO: 727458

Lat: 47.211N

Lon: 93.510W

Elev: 1355

StdP: 13.99

Time Zone: -6.00 (NAC)

Period: 94-19

WBAN: 94919

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-19.6	-15.1	-26.9	1.3	-15.8	-23.9	1.5	-12.5	24.8	10.8	22.8	11.4	5.5	310	0.612

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	20.2	84.2	69.8	81.6	67.6	79.3	65.7	72.5	81.6	70.1	79.1	68.1	76.6	10.5	180

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
69.7	114.9	79.5	66.4	102.2	75.6	64.3	95.0	73.3	37.0	81.6	34.9	79.3	33.2	76.5	81.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 21.7	19.2	17.3	DB	-25.5	89.4	6.1	2.7	-29.8	91.4	-33.4	92.9	-36.8	94.4	-41.2	96.4	(4)
(5)			WB	-24.3	75.9	4.3	2.1	-27.4	77.4	-30.0	78.7	-32.4	79.9	-35.6	81.4	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	40.2	9.5	13.7	26.7	39.8	52.2	62.1	67.1	64.7	56.9	44.0	28.6	15.6		
	DBStd	22.33	13.64	13.05	12.62	9.52	8.11	6.56	5.59	5.53	8.22	8.70	10.78	12.40		
	HDD50	5361	1255	1017	725	324	69	3	0	0	28	230	644	1066		
	HDD65	9289	1720	1437	1187	755	406	128	43	74	263	652	1093	1531		
	CDD50	1788	0	0	3	19	136	365	529	455	236	43	2	0		
	CDD65	242	0	0	0	0	8	41	107	64	21	1	0	0		
	CDH74	2176	0	0	0	7	131	423	929	507	166	13	0	0		
	CDH80	443	0	0	0	0	24	88	212	91	26	2	0	0		
	WSAvg	7.6	8.0	7.9	7.9	8.0	7.9	6.7	6.5	6.2	7.3	8.1	8.3	7.9		
	PrecAvg	27.4	0.9	0.5	1.3	2.1	3.1	4.1	3.9	3.6	3.3	2.3	1.3	1.0		
(15) Precipitation	PrecMax	38.0	3.1	1.5	3.0	4.3	6.4	7.5	9.4	10.0	7.5	6.8	3.1	2.8		
	PrecMin	18.0	0.1	0.0	0.2	0.3	0.4	0.5	0.8	0.6	0.6	0.4	0.1	0.2		
	PrecStd	4.5	0.7	0.4	0.7	1.1	1.4	1.7	1.9	2.4	1.9	1.8	0.9	0.5		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	42.7	46.0	63.5	73.0	83.1	87.6	89.9	87.8	82.4	75.0	60.6	43.3		
		MCWB	35.8	38.7	52.1	56.4	65.1	72.2	74.5	72.4	68.3	59.7	51.6	37.1		
	2%	DB	35.8	39.4	54.4	66.1	78.6	82.1	84.4	82.0	79.2	67.9	53.8	37.9		
		MCWB	32.5	34.6	45.0	50.7	60.7	67.4	70.8	68.4	66.1	56.9	46.2	34.0		
	5%	DB	31.9	36.4	48.0	62.7	72.8	79.1	81.9	79.5	74.7	62.9	47.2	34.6		
		MCWB	29.8	32.5	41.3	47.5	57.4	64.9	68.5	66.2	63.8	53.2	41.5	31.6		
	10%	DB	28.1	33.1	43.4	56.8	68.2	75.2	79.4	76.6	71.6	57.2	43.4	32.2		
		MCWB	26.0	29.9	37.0	44.5	54.7	62.4	66.8	64.9	61.8	49.5	38.1	30.3		
	0.4%	WB	36.5	39.7	55.9	58.0	68.1	73.9	76.0	73.8	71.3	62.0	52.6	38.3		
		MCDB	41.2	45.3	61.8	71.6	78.9	83.7	85.6	83.1	79.7	70.9	58.5	41.6		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	2%	WB	32.6	35.7	46.0	52.1	63.3	70.0	73.3	71.0	67.9	58.1	46.0	34.2		
		MCDB	35.1	39.6	52.4	63.6	73.1	80.2	82.5	80.0	76.1	66.2	51.0	37.8		
	5%	WB	29.6	32.4	41.5	48.6	59.4	67.4	70.7	68.5	65.7	54.3	42.0	32.1		
		MCDB	31.8	36.3	47.8	60.6	69.1	76.5	79.4	76.5	73.0	61.3	47.1	34.5		
	10%	WB	26.2	29.4	37.6	45.4	56.2	64.7	68.5	66.6	63.0	50.5	38.5	29.9		
		MCDB	28.1	32.5	43.2	55.5	66.3	72.8	77.1	74.3	69.5	56.8	43.2	32.0		
	5% DB	MDBR	17.4	18.8	18.7	20.1	21.7	20.6	20.2	19.8	19.2	15.9	13.8	14.4		
		MCDBR	17.4	18.8	23.1	29.7	29.2	25.1	23.0	23.1	23.8	24.1	20.1	16.2		
		MCWBR	14.7	14.7	15.6	17.9	15.8	13.8	12.4	12.3	13.9	14.8	14.3	13.6		
		MCWBR	16.4	17.6	21.7	26.6	24.3	21.8	21.2	20.2	20.1	21.5	18.6	14.9		
(40) Clear-Sky Solar Irradiance	taub		0.251	0.268	0.306	0.348	0.364	0.369	0.379	0.375	0.349	0.307	0.284	0.263		
		taud	2.377	2.334	2.381	2.373	2.380	2.389	2.379	2.416	2.466	2.574	2.520	2.402		
	Ebn at Noon		271	288	292	288	285	284	279	276	274	273	258	251		
		Edh at Noon	22	28	32	36	37	37	36	34	30	23	19	19		
(44) All-Sky Solar Radiation	RadAvg		464	823	1186	1474	1672	1795	1924	1623	1211	728	430	328		
	RadStd		53	75	112	147	122	132	84	123	92	99	52	33		

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	-2.77	N/A	N/A	N/A	+1.67	N/A	N/A	N/A	N/A
(47) Regional (38 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page